

DNA Packaging: Study Notes

Session Objectives:

SOM.MK.I.BPM1.1.FTM.3.BCHM.0028	Differentiate between a supercoiled and a relaxed DNA molecule
SOM.MK.I.BPM1.1.FTM.3.BCHM.0029	State the reasons why supercoiling of DNA molecules is important.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0030	Compare and contrast DNA packaging in prokaryotes and eukaryotes.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0031	Describe the structure of chromatin and histone proteins.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0032	Explain how dsDNA and histones associate to form the nucleosome.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0033	Describe the different structural conformations of DNA packaging in eukaryotes (10 nm Fiber , 30 nm fiber and metaphase chromosomes).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0034	Describe how histone modification can influence transcriptional activity

Assigned Reading: Lippincotts Illustrated Review Biochemistry.

<http://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073> **Chapter 30. DNA Structure, Replication, and Repair | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library (lwwhealthlibrary.com):** V. Eukaryotic DNA Organization

IMPORTANT: Please see Panopto videos on the topic DNA packaging.

The DNA molecules in both prokaryote and eukaryotes are large relative the structure in which they are found. As a result, it is necessary for these molecules to be packaged in such a way that they can physically fit into the cell (prokaryotes) or the nucleus (eukaryotes). Packaging DNA typically involves a process called *supercoiling* as well as the association of various proteins with the DNA.

Supercoiling

Many DNA molecules are circular and therefore have no free ends. Some examples include bacterial chromosomes, bacterial plasmids and the mitochondrial DNA.

Close, circular DNA can exist in forms whereby the helix has been twisted about its own axis, either with the helix (to the right) that produces positive supercoils or against it (to the left) producing negative supercoils. DNA molecules that have these additional twists applied to them tend to coil up upon themselves and are therefore termed, supercoiled.

The phenomenon can be easily demonstrated using a rubber band or the coiled cord of a phone receiver. The addition of twists to these examples induces the formation of additional coils, and the rubber band or phone cord begins to wrap around itself.

These changes in the DNA helix can produce molecules of identical sequence but which possess different topology. These DNA molecules are called topological isomers.

A DNA molecule that contains no supercoils is said to be relaxed. Energy is required in order for this molecule to become supercoiled. Proteins and enzymes that interact with the DNA provide the energy necessary to induce supercoiling. Some of the enzymes act directly on the DNA to induce twisting, while other proteins such as DNA polymerase or RNA polymerase induce supercoiling during the processes of DNA replication or transcription. Potential energy is stored in the supercoiling of the DNA, therefore as more supercoiling is induced in a DNA molecule the more energy is required to induce further supercoiling. In other words, as the molecule becomes more twisted it gets harder to twist the molecule.

Because a DNA molecule with a free end can rotate freely about its axis, supercoils can only exist in double-helical DNA if the ends of the DNA are not free. In prokaryote DNA this normally occurs, because the two ends of the DNA molecule are joined together to make a circle. Despite the fact that eukaryotic chromosomes are linear and possess free ends supercoiling still occurs in eukaryotes. This is because the very long chromosomes of eukaryotes are attached to various nuclear and scaffold proteins that tend to make portions of the chromosome immobile.

Similarly, if one or both ends of a supercoiled DNA molecule are freed, or if a single strand or both strands of a supercoiled DNA is cleaved either deliberately or by accident, untwisting can occur and the supercoils are lost.

Supercoiling in DNA is important for 3 reasons:

- 1) Supercoiled DNA molecules are more compact than relaxed DNA molecules and therefore take up less space. As result the DNA is able to fit into the confined space of the cell or nucleus.
- 2) Negative supercoiling promotes strand separation in the double helix. This negative supercoiling can assist in facilitating DNA strand separation during DNA replication and transcription. In the case of prokaryotes, the DNA exists naturally in a negatively supercoiled state.
- 3) Positive supercoiling makes DNA strand separation more difficult, and this is a problem when strand separation is necessary – for example – during DNA replication and transcription. Because of this, positive supercoiling is not a natural state of DNA. Positive supercoiling are however introduced into DNA molecules during the unwinding process that occurs with DNA replication and transcription. This positive supercoiling that is introduced by the activities of DNA polymerase and RNA polymerase can be remedied by the introduction of negative supercoils. Indeed, for every 10 base pairs that are replicated during DNA replication there is a corresponding introduction of a positive supercoil ahead of the DNA polymerase. These positive supercoils must be corrected, by

the introduction of a negative supercoils, in order for DNA replication to continue efficiently.

DNA Topoisomerases

Enzymes exist that can introduce or remove supercoils into relaxed DNA. These enzymes are called topoisomerases. In addition, these enzymes aid in preventing tangling of DNA. In the cases where supercoils are being added to the DNA there is an energy requirement, and this is provided by the hydrolysis of ATP.

There are two classes of topoisomerases:

Type I Topoisomerases

Type I topoisomerases produce a transient single-stranded break or nick in only one strand of the DNA helix. Topoisomerase I does this by forming a covalent linkage with the DNA phosphate group. The break in the phosphodiester bond of the backbone allows the two sections of DNA on either side of the nick to rotate freely relative to each other. The intact phosphodiester bond in this case acts as a swivel point for this rotation. The existing supercoiling on the DNA helix will drive the rotation on the molecule in the direction that will relieve the tension. Once the tension is relieved the covalent bond between topoisomerase I and the phosphate is broken and the phosphodiester bond is restored. As a result of this, tensions in the DNA helix ahead of DNA polymerase or RNA polymerase is reduced and DNA replication and transcription can progress.

Type II Topoisomerases

Type II topoisomerases act by producing transient breaks in both strands of the DNA helix. It does this by forming a covalent linkage with both strands of the DNA helix at the same time. This enzyme is activated at sites on chromosomes where two DNA helices cross over each other (ie. potential for tangling). The topoisomerase II binds to the site where the DNA molecules are crossing each other and using ATP hydrolysis carries out the following reaction:

- 1) it breaks the DNA helix at both strands while maintaining attachment to both free ends of the DNA. This produces a “gate” for which another DNA helix can pass.
- 2) the second DNA helix passes through the “gate”
- 3) it reseals the break and disassociates from the DNA.

One of the best understood type II topoisomerases in prokaryotes is DNA gyrase. The quinolone family of antibiotics, including nalidixic acid and ciprofloxacin, work by inhibiting DNA gyrase. As a result, this causes a halt in DNA replication due to the build up of positive supercoiling ahead of the replication fork during DNA replication.

DNA Packaging in Eukaryotes

Each eukaryote chromosome is composed of a single DNA molecule. These chromosomes are generally much larger than the DNA molecules found in prokaryotes. As a result, the problem of packaging DNA in eukaryotes is more significant than that found in prokaryotes. The DNA must be stored in a compact form to protect it from physical damage such as shearing, and to prevent the tangling and knotting of DNA with the DNA of other chromosomes. This is particularly important during cell division.

The way in which eukaryotes package their DNA is different compared to packaging in prokaryotes. In eukaryote cells, except sperm, DNA is tightly associated with a group of small, basic proteins called histones. The histones are rich in the amino acids lysine and arginine, and as a result generally carry a positive charge. The negatively charged DNA will therefore readily associate with histone proteins.

The association of DNA with histones is called chromatin. Indeed, approximately half of the mass of chromatin is composed of histone proteins.

In sperm cells DNA is associated with a different class of basic proteins called protamines. These proteins serve a similar function as histones but facilitate even tighter packaging of the DNA. Following fertilization, the protamines are removed from the paternal DNA and are replaced by histone proteins.

There are five types of histone proteins: H1, H2A, H2B, H3 and H4. Each of these proteins is found in equimolar amounts in the cell with the exception of H1 which is present in half the molar amount of the other histones.

The Nucleosome

Under certain conditions, when chromatin is examined under the electron microscope it appears as a single fibre connecting bead-like particles – ie. beads on a string. The particles or beads are called nucleosomes.

The nucleosome is a disc-shaped histone core structure around which approximately 160-200 base pairs (1.65 turns) of DNA is wrapped. The nucleosome core (octamer) is composed of two molecules each of H2A, H2B, H3 and H4 histones.

The DNA that is wrapped around the nucleosome core is partially held in place by another histone protein that is not part of the nucleosome core. The histone is H1.

Varying lengths of DNA that is called “linker DNA” separates each “bead”.

The DNA is about 7 times more compact in the nucleosome than when it is not associated with histone proteins.

The Solenoid Structure

In order to further package the DNA there are interactions between adjacent nucleosomes. The more compacted structure is termed the solenoid structure. Under high ionic strength, when chromatin is observed under the electron microscope it does not appear as “beads on a string”. Instead, we observe thicker fibers with a diameter of approximately 30 nm.

This solenoid structure is formed in part by interactions between adjacent H1 histones. This produces a packing ratio of approximately 1:40.

Further compaction (5-10 fold) occurs as a result of supercoiling and other protein interactions to produce the very compacted metaphase chromosome.

Modification of Histones

The amino acid residues found in the tail portion of the histone proteins, especially serine, lysine and arginine, are subject to a number of different covalent modifications. These modifications include acetylation, phosphorylation, methylation and ubiquitination. Each of these modifications can affect the level of association of the DNA with the nucleosome, and as a result affect DNA compaction. These modifications in turn affect the transcriptional activity of the DNA. Highly compacted DNA, termed heterochromatin, is transcriptionally inactive. Loosely compacted DNA, termed euchromatin, is transcriptionally active.

The relationship between histone modifications and transcriptional activity of the associated DNA is complex.

In the case of histone acetylation and deacetylation, there is a clear relationship between these modifications and gene activity. Genes associated with acetylated histones are typically transcriptionally active, while genes associated with deacetylated histones are typically transcriptionally inactive. This process of histone acetylation/deacetylation is regulated and initiated by modifications to the bases of the DNA itself – in particular cytosine methylation.

In the case of the other modifications such as phosphorylation, methylation and ubiquitination the relationship between the modification and gene expression is not as clear. Various combinations and permutations of these modifications can result in either gene expression or gene silencing. This has led to what is called the histone code hypothesis.